The error of the Guy–Kelly heuristic, measured against exact counts

Aleksei Kudriashov (Alex Komang)
Nusa Dua Studio, Nusa Dua, Bali, Indonesia
ORCID: 0009-0006-8885-5338

21 August 2026 — version 1.0

Abstract

The Guy–Kelly heuristic for the no-three-in-line problem is a first-moment threshold: one equates $\log \binom{n^2}{m}$ with the expected number of collinear triples in a random m-subset, computed as if the triples were independent. Its corrected constant is $\pi/\sqrt{3} = 1.8138$ (Guy, after Ellmann, 2004). We test the heuristic not at its threshold but on the quantity it actually predicts — the *number* of configurations — against exact counts, which are available for $m = 2n$ up to $n = 19$ (A000755, and $n = 20$ from Flammenkamp's table [5]) and, at the ratio $m/2n = 0.9$, up to $n = 20$ by exhaustive enumeration and an unbiased tree-size estimator validated against it.

Three things come out. The heuristic overestimates the count of $2n$-point configurations by $10^{6.9}$ at $n = 20$, but the per-step growth of that error *decays*. Refining the heuristic — treating lines rather than triples as independent — makes it *twice as wrong*, because the crude version carries two errors of opposite sign that partly cancel; we measure the cancellation at $10^{2.65}$. And at the fixed ratio $m/2n = 0.9$, away from the hard ceiling $2n$, the multiplicative error is bounded on $n = 5, \dots, 20$: continued growth at the initial rate is excluded at thirteen standard deviations. A bounded multiplier does not move an exponential rate, so nothing in these data argues against $\pi/\sqrt{3}$.

1 The heuristic, and which constant is which

Let $T(n)$ be the number of collinear triples in the $n \times n$ grid. It has a closed form,

$$T(n) = \sum_{\gcd(a,b)=1} \sum_{s \geq 2} (s-1)(n - s|a|)^+ (n - s|b|)^+,$$

which we verified against brute force on eighteen values. The heuristic estimates the number of admissible m-subsets by

$$\binom{N}{m} \exp\Big(-T(n) \binom{m}{3} \Big/ \binom{N}{3} \Big), \qquad N = n^2, \tag{1}$$

and locates the threshold where this drops below one. Guy and Kelly's original constant was $(2\pi^2/3)^{1/3} = 1.8739$; Gabor Ellmann found an error in the argument in March 2004, and the corrected value, recorded by Guy in OEIS A000769 that October and documented by Voutier (arXiv:2603.00215), is

$$c = \pi/\sqrt{3} = 1.813799.$$

Our own derivation of the threshold reproduces the corrected value, not the retracted one. As a check of the machinery we located the crossing exactly: the threshold from (1) first falls below $2n$ at $n = 493$ (at $n = 492$ it equals $984 = 2n$; at $n = 493$ it is $985 < 986$), which agrees to the integer with the value reported by Prellberg [3] and quoted in the recent survey [4]: "the heuristic probabilistic argument only applies for $n \geq 493$". Our computation is independent of his and reproduces the same integer.

2 Testing the count, not the threshold

A threshold inherits the whole error of the estimate without exhibiting its size. The count does not. Exact counts of $2n$-point configurations are A000755, published to $n = 19$; the value at $n = 20$ is on A. Flammenkamp's table. We obtain both independently by summing orbit sizes $8/|\text{stab}|$ over the D_4-classes of his solution database, which reproduces $a(2), \ldots, a(19)$ exactly and gives 941580 at $n = 20$. Against (1):

n	exact	estimate	$\log_{10}$(exact/estimate)
6	50	$10^{4.12}$	-2.421
10	1135	$10^{7.11}$	-4.055
14	10568	$10^{9.63}$	-5.604
18	152210	$10^{11.78}$	-6.597
20	941580	$10^{12.87}$	-6.896

The error reaches seven orders of magnitude. Its increment per step, however, decays by a factor of four across the range: $-0.86, \ldots, -0.25, -0.13$. That distinction — between an error whose increment is constant and one whose increment decays — is what the rest of this note turns on.

3 Refining the heuristic makes it worse

Triples on one line are strongly dependent. The natural repair is to treat *lines* as independent and use the exact hypergeometric probability that a line of k grid points carries at most two chosen points. It is worse, and by a clean factor:

	triples independent	lines independent
$n = 6$, $m = 12$	-2.421	-4.107
$n = 7$, $m = 14$	-2.639	-5.285

The mechanism is that (1) carries *two* errors of opposite sign. Independence *between* lines overestimates, since a set that has already avoided one line avoids the next more easily. Counting the $\binom{k}{3}$ overlapping triples on a line as independent events overestimates the probability of a violation, hence underestimates the survival probability. Replacing the second by an exact computation removes the compensation and leaves the first bare. At $n = 7$ the cancellation is worth $10^{2.65}$.

The practical corollary is uncomfortable: any refinement that fixes one of the two errors will make the heuristic worse. Only fixing both helps, and the first requires the correlations between lines, which is precisely what nobody knows how to compute. This may be why the 1968 form has not been superseded in fifty-eight years.

The same mechanism predicts where the heuristic should fail hardest, and we can test that prediction on our own data. In a symmetric class the points are rigidly coupled, so the first error grows while the second does not, and the cancellation should break. We measured the coupling directly: among all maximum configurations (complete enumeration, $n \leq 11$), those carrying a symmetry of the square are over-represented relative to a null model matched on row and column counts by factors $\times 8, \times 56, \times 125, \times 769, \times 3729$ at $n = 6, \ldots, 10$. Where the enrichment is three orders of magnitude, independence between constraints is not a small approximation, and a heuristic built on it should not be trusted — as reportedly it is not, for symmetric classes.

4 Which error is which class

Over sixteen steps the line-independent version has an essentially constant increment, -1.6 per step, while the triple-independent version decays fourfold. A constant increment means an error growing exponentially in n, i.e. a wrong exponent; a decaying increment means a wrong multiplier only. So the refinement is not merely further from the truth on a given n: it is wrong in a different and worse way.

5 At fixed ratio, the multiplier is bounded

All of the above is at $m = 2n$, which is the hard ceiling (two points per row). The decay of the increment could be an artefact of approaching that ceiling rather than a property of the exponent. We therefore fixed the ratio $r = m/2n = 0.9$ and measured the error as n grows. Two of the four points are exhaustive enumerations; the other two use an unbiased tree-size estimator of Knuth type, validated against exact counts at $n = 6, \dots, 10$ with errors 0.35% to 2.14%, and combined across independent seeds in the arithmetic mean, in which the estimator is unbiased.

n	m	count	error	increment per unit n
5	9	840 (exact)	-0.975	
10	18	92 734 158 (exact)	-1.771	-0.1592
15	27	$3.70 \cdot 10^{13}$	-1.743 ± 0.100	$+0.0056 \pm 0.0199$
20	36	$2.03 \cdot 10^{19}$	-1.431 ± 0.144	$+0.0625 \pm 0.0351$

Had the initial rate persisted, the error at $n = 20$ would be -3.363; it is -1.431, a discrepancy of 1.933 against an uncertainty of 0.144 — thirteen standard deviations. The multiplicative error is bounded on this range, and if it moves at all it shrinks. A bounded multiplier leaves the exponential rate untouched, and the rate is the constant.

We do not claim more. Whether the plateau is flat or slowly shrinking is unresolved: the last increment is $+0.0625 \pm 0.0351$, under two standard deviations from zero. Beyond $n = 20$ the estimator's random descents no longer reach the required depth — at $n = 25$, zero hits in two million — so $n \leq 20$ is the reach of this method.

6 On three dimensions

The same test was carried out in the cube (no four coplanar) on complete enumerations at $n = 3, 4, 5$ by the first solver. There the increment of the error *grows*: $+2.22$, then $+5.53$. By the criterion above that is a wrong exponent. The conclusion of Section 5 is therefore two-dimensional and must not be carried into the cube. This is consistent with the mechanism: in the plane the hard ceiling $2n$ binds and anchors the estimate, while in the cube the corresponding bound $3n$ stands far from the achievable values and does not.

References

[1] R. K. Guy and P. A. Kelly, The no-three-in-line problem, *Canad. Math. Bull.* 11 (1968) 527–531.

[2] P. M. Voutier, On the Guy–Kelly conjecture for the no-three-in-line problem, arXiv:2603.00215 (2026).

[3] T. Prellberg, Constraint satisfaction programming for the no-three-in-line problem, *J. Combin. Theory* 225 (2027) 106244.

[4] The general position problem: a survey, arXiv:2501.19385 (2025).

[5] A. Flammenkamp, The no-three-in-line problem, `https://wwwhomes.uni-bielefeld.de/achim/no3in/readme.html`.

AI/tool-use disclosure

The computations, the programs and the text were produced by autonomous AI agents (Anthropic Claude) under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Two agents worked independently, one in each dimension, and each checked the other's conclusions rather than its code. Programs, journals and the full record — including the measurements that refuted our own earlier claims — are at `https://github.com/iwasborninbali/saturation`.